

HYPERBOLIC FUNCTIONS



SYNOPSIS

- If x is any real number then

$e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n} + \dots \infty$ is called the exponential series.

- $e^{-x} = 1 - \frac{x}{1} + \frac{x^2}{2} - \frac{x^3}{3} + \dots + (-1)^n \frac{x^n}{n} + \dots \infty$.

- The function e^x can be written as

$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \text{ for all } x \in \mathbb{R};$$

- (i) $\sinh x = \frac{e^x - e^{-x}}{2}$
- $$= \frac{x}{1} + \frac{x^3}{3} + \frac{x^5}{5} + \dots, x \in \mathbb{R}$$

(ii) $\cosh x = \frac{e^x + e^{-x}}{2}$

$$= 1 + \frac{x^2}{2} + \frac{x^4}{4} + \dots, x \in \mathbb{R}$$

(iii) $\tanh x = \frac{\sinh x}{\cosh x}, x \in \mathbb{R}$

(iv) $\operatorname{sech} x = \frac{1}{\cosh x}, x \in \mathbb{R}$

(v) $\operatorname{cosech} x = \frac{1}{\sinh x}, x \in \mathbb{R} - \{0\}$

(vi) $\coth x = \frac{\cosh x}{\sinh x}, x \in \mathbb{R} - \{0\}$

- Hyperbolic functions are not circular functions and hence are not meant to use trigonometric identities.

- **Hyperbolic Identities:**

(i) $\cosh^2 x - \sinh^2 x = 1, \forall x \in \mathbb{R}$

(ii) $\operatorname{sech}^2 x + \tanh^2 x = 1, \forall x \in \mathbb{R}$

(iii) $\coth^2 x - \operatorname{cosech}^2 x = 1, \forall x \in \mathbb{R} - \{0\}$

W.E-1: $\sinh x = 3/4$ then $\cosh x =$

Sol: $\cosh^2 x - \sinh^2 x = 1 \Rightarrow \cosh^2 x = 1 + \sinh^2 x$

$$= 1 + \frac{9}{16} \Rightarrow \cosh x = \frac{5}{4}$$

- **Properties of Hyperbolic Functions:**

(i) $\sinh(-x) = -\sinh x$ (ii) $\cosh(-x) = \cosh x$

(iii) $\tanh(-x) = -\tanh x$

(iv) $\coth(-x) = -\coth x$ (v) $\operatorname{sech}(-x) = \operatorname{sech} x$

(vi) $\operatorname{cosech}(-x) = -\operatorname{cosech} x$

- (i) $\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$

(ii) $\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

(iii) $\tanh(x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$

(iv) $\coth(x \pm y) = \frac{\coth x \coth y \pm 1}{\coth y \pm \coth x}$

- (i) $\sinh 2x = 2 \sinh x \cosh x = \frac{2 \tanh x}{1 - \tanh^2 x}$

(ii) $\cosh 2x = \cosh^2 x + \sinh^2 x = \frac{1 + \tanh^2 x}{1 - \tanh^2 x}$

(iii) $\tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$

(iv) $\sinh 2x + \cosh 2x = \frac{1 + \tanh x}{1 - \tanh x}$

W.E-2: $\tanh x = 3/5$ then $\cosh(2x) =$

Sol: $\cosh 2x = \frac{1 + \tanh^2 x}{1 - \tanh^2 x} = \frac{34}{16} = \frac{17}{8}$

- (i) $\sinh 3x = 3 \sinh x + 4 \sinh^3 x$

(ii) $\cosh 3x = 4 \cosh^3 x - 3 \cosh x$

(iii) $\tanh(3x) = \frac{3 \tanh x + \tanh^3 x}{1 + 3 \tanh^2 x}$

W.E-3: $\tanh x = 1/2$ then $\tanh(3x) =$

Sol: $\tanh(3x) = \frac{3 \tanh(x) + \tanh^3(x)}{1 + 3 \tanh^2(x)} = \frac{13}{14}$

- $\sinh(x+y) \sinh(x-y) = \sinh^2 x - \sinh^2 y$
- $\cosh(x+y) \cosh(x-y) = \cosh^2 x + \sinh^2 y$
- $(\cosh x + \sinh x)^n = (\cosh(nx) + \sinh(nx)) = e^{nx}$
- $(\cosh x - \sinh x)^n = (\cosh(nx) - \sinh(nx)) = e^{-nx}$

➤ $\cosh 2nx + \sinh 2nx = \left(\frac{1 + \tanh x}{1 - \tanh x} \right)^n$

- $\cosh x$, $\operatorname{sech} x$ are even functions.
- $\sinh x$, $\operatorname{cosech} x$, $\tanh x$ and $\coth x$ are odd functions.

W.E.4: Show that $f(x) = \cosh x$ is an even function

Sol: $f(-x) = \cosh(-x) = \frac{e^{-x} + e^{(-x)}}{2} = \frac{e^{-x} + e^x}{2} = f(x)$

- None of the six hyperbolic functions are periodic.
- $\sinh x$, $\tanh x$, $\coth x$ and $\operatorname{cosech} x$ are one-one functions but $\cosh x$ and $\operatorname{sech} x$ are not one-one functions as

$\cosh(-x) = \cosh x$ and $\operatorname{sech}(-x) = \operatorname{sech} x$
for all $x \in \mathbb{R}$

- $\sinh x + \cosh x$, $\sinh x - \cosh x$ are bijective functions

- **Domain and range of hyperbolic functions:**

Function	Domain	Range
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$
$\cosh x$	$\mathbb{R}$	$[1, \infty)$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$
$\coth x$	$\mathbb{R} - \{0\}$	$\mathbb{R} - [-1, 1]$
$\operatorname{cosech} x$	$\mathbb{R} - \{0\}$	$\mathbb{R} - \{0\}$
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$

- Since hyperbolic functions are defined in terms of exponential functions. Therefore Inverse hyperbolic functions can be expressed in terms of logarithmic functions.

- **Inverse Hyperbolic Functions in Terms of Logarithmic Functions:**

i) $\sinh^{-1} x = \log_e \left(x + \sqrt{x^2 + 1} \right) \forall x \in \mathbb{R}$

W.E-5: If $\sinh x = 3$. Then $x = \dots$

Sol: $\sinh x = 3 \Rightarrow x = \sinh^{-1}(3) = \log \left(3 + \sqrt{3^2 + 1} \right)$
 $= \log \left(3 + \sqrt{10} \right)$

ii) $\cosh^{-1}(x) = \log_e \left(x + \sqrt{x^2 - 1} \right)$ for $x \geq 1$

W.E - 6: If $\cosh x = 3$ then $x = \dots$

Sol: $\cosh x = 3 \Rightarrow x = \cosh^{-1}(3) = \log \left(3 + \sqrt{3^2 - 1} \right) = \log \left(3 + \sqrt{8} \right)$

iii) $\tanh^{-1}(x) = \frac{1}{2} \log \left(\frac{1+x}{1-x} \right)$ for $x \in (-1, 1)$

Similarly we have

iv) $\coth^{-1}(x) = \frac{1}{2} \log \left(\frac{x+1}{x-1} \right)$ for $|x| > 1$

v) $\operatorname{sech}^{-1}(x) = \log_e \left(\frac{1 + \sqrt{1-x^2}}{x} \right)$ for $x \in (0, 1]$

vi) $\operatorname{cosech}^{-1}(x) = \log_e \left(\frac{1 + \sqrt{1+x^2}}{x} \right)$ for $x > 0$
 $= \log_e \left[\frac{1 - \sqrt{1+x^2}}{x} \right]$ for $x < 0$

- **Domain and Range of inverse Hyperbolic Functions:**

Function	Domain	Range
$\sinh^{-1} x$	$\mathbb{R}$	$\mathbb{R}$
$\cosh^{-1} x$	$[1, \infty)$	$[0, \infty)$
$\tanh^{-1} x$	$(-1, 1)$	$\mathbb{R}$
$\coth^{-1} x$	$\mathbb{R} - [-1, 1]$	$\mathbb{R} - \{0\}$
$\operatorname{sech}^{-1} x$	$(0, 1]$	$[0, \infty)$
$\operatorname{cosech}^{-1} x$	$\mathbb{R} - \{0\}$	$\mathbb{R} - \{0\}$

➤ $\sinh^{-1}(x) = \cosh^{-1} \left(\sqrt{x^2 + 1} \right)$
 $= \operatorname{cosech}^{-1} \left(\frac{1}{x} \right)$
 $= \tanh^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$

$$\begin{aligned} \Rightarrow \cosh^{-1}(x) &= \sinh^{-1}(\sqrt{x^2 - 1}) \\ &= \sec h^{-1}\left(\frac{1}{x}\right) = \tanh^{-1}\left(\frac{\sqrt{x^2 - 1}}{x}\right) \end{aligned}$$

➤ **Euler's Formula :**

$$e^{ix} = \cos x + i \sin x \quad \forall x \in R$$

$$e^{-ix} = \cos x - i \sin x \quad \forall x \in R$$

➤ **Hyperbolic Functions Using Euler's Formula :**

$$\sinh(ix) = i \sin x$$

$$\cosh(ix) = \cos x$$

$$\tanh(ix) = i \tan x$$

$$\coth(ix) = -i \cot x$$

$$\operatorname{cosech}(ix) = -i \operatorname{cosec} x$$

$$\operatorname{sech}(ix) = \sec x$$

$$\Rightarrow \sin ix = i \sinh x, \cos ix = \cosh x$$

$$\tan ix = i \tanh x$$

$$\cot ix = -i \coth x, \sec ix = \sec hx$$

$$\operatorname{cosec} ix = -i \operatorname{cosec} hx$$

$$\Rightarrow \sinh^{-1} x = -i \sin^{-1} ix$$

$$\cosh^{-1} x = -i \cos^{-1} x$$

$$\tanh^{-1} x = -i \tan^{-1} ix$$



LEVEL - I (C.W)

1. The domain of $\operatorname{cosech} x$ is

1) $(-\infty, \infty)$ 2) $(-\infty, 0) \cup (0, \infty)$

3) $(0, \infty)$ 4) $(-\infty, 0)$

2. The range of $\coth x$ is

1) $(-\infty, -1) \cup (1, \infty)$ 2) $[-1, 1]$

3) $(-\infty, \infty)$ 4) $(-\infty, -1)$

3. $\cosh(0) =$

1) 0 2) 1 3) e 4) not defined

4. $\cosh(2) + \sinh(2) =$ (EAM-2000)

1) $\frac{1}{e}$ 2) e 3) $\frac{1}{e^2}$ 4) e^2

5. $\sinh(3) - \cosh(3) =$

1) e^{-3} 2) $-e^{-3}$ 3) e^3 4) $-e^3$

6. $(\cosh 2 + \sinh 2)^n =$

1) $\cosh(2n) + \sinh(2n)$

2) $\cosh(2n) - \sinh(2n)$

3) $\cosh(2n)$ 4) $\sinh(2n)$

7. If $\cosh x = \sec \alpha$ then $\operatorname{cosec} hx =$

1) $\sin \alpha$ 2) $\cos \alpha$

3) $\tan \alpha$ 4) $\cot \alpha$

8. If $\sinh x = \frac{3}{4}$ then $\sinh(2x) =$

1) $\frac{5}{8}$ 2) $\frac{15}{8}$ 3) $\frac{7}{8}$ 4) $\frac{17}{8}$

9. If $\tanh x = \frac{3}{5}$ then $\sinh(2x) =$

1) $\frac{15}{8}$ 2) $\frac{15}{17}$ 3) $\frac{18}{17}$ 4) $\frac{17}{8}$

10. $\cosh^4(x) - \sinh^4(x) =$

1) $\cosh x$ 2) $\cosh 2x$ 3) $\sinh x$ 4) $\sinh 2x$

11. If $\cosh x = \frac{5}{4}$ then $\cosh(3x) =$

1) $\frac{61}{16}$ 2) $\frac{63}{16}$ 3) $\frac{61}{63}$ 4) $\frac{65}{16}$

12. $\sinh^{-1}\left(2^{\frac{3}{2}}\right) =$

1) $\log_e(2 + \sqrt{18})$ 2) $\log_e(3 + \sqrt{8})$

3) $\log_e(3 - \sqrt{8})$ 4) $\log_e(\sqrt{8} + \sqrt{27})$

13. $\cosh^{-1}(2) =$

1) $\log_e(2 + \sqrt{3})$ 2) $\log_e(2 + \sqrt{5})$

3) $\log_e(2 - \sqrt{5})$ 4) $\log_e(2 + \sqrt{2})$

14. $\tanh^{-1}\left(\frac{1}{3}\right) =$

1) $\frac{1}{2} \log_e(2)$ 2) $2 \log_e(2)$

- 3) $-\frac{1}{2}\log_e(2)$ 4) $-2\log_e(2)$
15. $2 \tanh^{-1}\left(\frac{1}{2}\right) =$
 1) 0 2) $\log_e 2$ 3) $\log_e 3$ 4) $\log_e 4$
16. $\operatorname{cosech}^{-1}(-4) =$
 1) $\log_e\left(\frac{\sqrt{17}-1}{4}\right)$ 2) $\log_e(\sqrt{17}-1)$
 3) $\log_e(\sqrt{17}+1)$ 4) $\log_e\left(\frac{\sqrt{17}+1}{4}\right)$
17. If $\sinh^{-1}(x) = \log_e(5 + \sqrt{26})$ then $x =$
 1) 1 2) 2 3) 3 4) 5
18. If $x = \tanh^{-1}(y)$ then $\log_e\left(\frac{1+y}{1-y}\right)$
 (EAM-1998)
 1) x 2) $4x$ 3) $2x$ 4) $3x$
19. If $\tanh^{-1}(x) = a \log_e\left(\frac{1+x}{1-x}\right), |x| < 1$ then $a =$
 (EAM-2010)
 1) 1 2) 2 3) $\frac{1}{2}$ 4) $\frac{1}{4}$
20. If $\cosh^{-1}(k) = \log_e(3 + 2\sqrt{2})$ then $k =$
 1) 1 2) 2 3) 3 4) 4
21. $e^{\sinh^{-1}(\cot \theta)} =$
 1) $\cot \theta + \operatorname{cosec} \theta$ 2) $-\cot \theta + \operatorname{cosec} \theta$
 3) $\sec \theta - \tan \theta$ 4) $\sec \theta + \tan \theta$
22. $\operatorname{sech}^{-1}(\sin \theta) =$
 1) $\log_e\left(\cos \frac{\theta}{2}\right)$ 2) $\log_e\left(\sin \frac{\theta}{2}\right)$
 3) $\log_e(\cos \theta)$ 4) $\log_e\left(\cot \frac{\theta}{2}\right)$
23. If $\sinh^{-1}(2) + \sinh^{-1}(3) = x$ then $\cosh(x) =$
 (EAM-2009)

- 1) $\frac{1}{2}(3\sqrt{5} + 2\sqrt{10})$ 2) $\frac{1}{2}(3\sqrt{5} - 2\sqrt{10})$
 3) $\frac{1}{2}(12 + 2\sqrt{50})$ 4) $\frac{1}{2}(12 - 2\sqrt{50})$
24. If $x = \log_e\left[\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)\right]$ then $\cosh x =$
 1) $\sec \theta$ 2) $\operatorname{cosec} \theta$ 3) $\sin \theta$ 4) $\cos \theta$
25. If $x = \log_e\left[\cot\left(\frac{\pi}{4} + \theta\right)\right]$ then $\sinh x =$
 1) $\tan 2\theta$ 2) $\cot 2\theta$
 3) $-\tan 2\theta$ 4) $-\cot 2\theta$
26. If $\cosh 2x = 99$ then $\tanh x =$
 1) $\frac{5}{7\sqrt{2}}$ 2) $\frac{7}{5\sqrt{2}}$ 3) $\frac{5\sqrt{7}}{2}$ 4) $\frac{7\sqrt{5}}{2}$
27. $\sinh(ix) =$
 1) $i \sin x$ 2) $\sin(ix)$ 3) $-i \sin x$ 4) $i \sin(ix)$

LEVEL - I (C.W)-KEY

- 01) 2 02) 1 03) 2 04) 4 05) 2
 06) 1 07) 4 08) 2 09) 1 10) 2
 11) 4 12) 2 13) 1 14) 1 15) 3
 16) 1 17) 4 18) 3 19) 3 20) 3
 21) 1 22) 4 23) 3 24) 1 25) 3
 26) 2 27) 1

LEVEL - I (C.W)-HINTS

- The domain of $\operatorname{cosech} x$ is $(-\infty, 0) \cup (0, \infty)$
- The range of $\coth x$ is $(-\infty, -1) \cup (1, \infty)$
- $\cosh(0) = 1$
- $\cosh(2) + \sinh(2) = e^2$
- $\frac{e^3 - e^{-3} - e^3 - e^{-3}}{2} = -e^{-3}$
- $(e^2)^n = \cosh(2n) + \sinh(2n)$
- $\cosh x = \sec \alpha$
 $\cosh^2 x - \sinh^2 x = 1$
 $\Rightarrow \sinh^2 x = \tan^2 \alpha$
 $\Rightarrow \operatorname{cosech} x = \cot \alpha$

8. $\sinh x = \frac{3}{4} \Rightarrow \cosh x = \frac{5}{4}$
 $\sinh(2x) = 2\left(\frac{3}{4}\right)\left(\frac{5}{4}\right)$
9. $\tanh x = \frac{3}{5} \Rightarrow \sinh 2x = \frac{2 \tanh x}{1 - \tanh^2 x}$
10. $\cosh^4 x - \sinh^4 x = \cosh 2x$
11. $\cosh 3x = 4 \cosh^3 x - 3 \cosh x$
12. $\sinh^{-1}(2\sqrt{2}) = \log(3 + \sqrt{8})$
13. $\cosh^{-1} 2 = \log(2 + \sqrt{3})$
14. $\tanh^{-1}\left(\frac{1}{3}\right) = \frac{1}{2} \log 2$
15. $2 \tanh^{-1}\left(\frac{1}{2}\right) = \log(3)$
16. $\operatorname{cosech}^{-1} x = \log\left(\frac{1 - \sqrt{1 + x^2}}{x}\right), \text{ for } x < 0$
17. $\sinh^{-1} x = \log(5 + \sqrt{5^2 + 1})$
 $\therefore x = 5$
18. $x = \tanh^{-1} y$
 $\Rightarrow x = \frac{1}{2} \log\left(\frac{1 + y}{1 - y}\right)$
19. $\tanh^{-1} x = a \log\left(\frac{1 + x}{1 - x}\right)$
 $\Rightarrow a = \frac{1}{2}$
20. $\cosh^{-1}(k) = \log(k + \sqrt{k^2 - 1})$
21. $e^{\log(\cot\theta + \sqrt{\cot^2\theta + 1})}$
 $= \cot\theta + \operatorname{cosec}\theta$
22. $\operatorname{sech}^{-1}(\sin\theta)$
 $= \log\left(\frac{1 + \sqrt{1 - \sin^2\theta}}{\sin\theta}\right)$

23. $\log(2 + \sqrt{5}) + \log(3 + \sqrt{10}) = x$
 $\Rightarrow e^x = 6 + 2\sqrt{10} + 3\sqrt{5} + \sqrt{50}$
24. $e^x = \frac{1 + \tan\theta/2}{1 - \tan\theta/2} \Rightarrow \cosh x = \sec\theta$
25. $e^x = \frac{1 - \tan\theta}{1 + \tan\theta}$
26. $\cosh 2x = 99$
 $\Rightarrow \frac{1 + \tanh^2\theta}{1 - \tanh^2\theta} = 99 \Rightarrow \tanh\theta = \frac{7}{5\sqrt{2}}$
27. $\sinh(ix) = i \sin x$

 LEVEL - I (H.W)

1. The domain of $\coth x$ is
 1) $(-\infty, \infty)$ 2) $(0, \infty)$ 3) $\mathbb{R} - \{0\}$ 4) $(-\infty, 0)$
2. The range of $\tanh x$ is
 1) $[-1, 1]$ 2) $(-1, 0)$ 3) $(0, 1)$ 4) $(-1, 1)$
3. $\sinh(0) =$
 1) 0 2) 1 3) e 4) not defined
4. $\cosh(1) - \sinh(1)$
 1) e^2 2) e 3) $\frac{1}{e}$ 4) $\frac{1}{e^2}$
5. $(\cosh x - \sinh x)^n =$
 1) $\cosh nx - \sinh nx$ 2) $2 \cosh nx$
 3) $2 \sinh nx$ 4) $\cosh nx + \sinh nx$
6. If $\cosh x = \frac{5}{3}$ then $\sinh x =$
 1) $\frac{3}{4}$ 2) $\frac{4}{3}$ 3) $\frac{4}{5}$ 4) $\frac{3}{5}$
7. If $\cosh x = \frac{5}{2}$ then $\cosh(2x) =$
 1) $\frac{5\sqrt{21}}{2}$ 2) $\frac{23}{5\sqrt{21}}$ 3) $\frac{23}{2}$ 4) $\frac{7}{2}$
8. If $\operatorname{sech} x = \frac{3}{5}$ then $\tanh(2x) =$

- 1) $\frac{40}{41}$ 2) $\frac{41}{40}$ 3) $\frac{21}{41}$ 4) $\frac{16}{41}$
9. $\cosh^4(2) - \sinh^4(2) =$
 1) $\cosh(2)$ 2) $\cosh(3)$
 3) $\cosh(4)$ 4) $\cosh(8)$
10. If $\sinh x = \frac{3}{4}$ then $\sinh(3x) =$
 1) $\frac{61}{16}$ 2) $\frac{63}{16}$ 3) $\frac{61}{63}$ 4) $\frac{65}{16}$
11. $\sinh^{-1}(2) =$
 1) $\log_e(2 + \sqrt{5})$ 2) $\log_e(2 + \sqrt{7})$
 3) $\log_e(3 + \sqrt{10})$ 4) $\log_e(5 + \sqrt{26})$
12. $\cosh^{-1}(1) =$
 1) 2 2) 3 3) $\log_e(1 + \sqrt{2})$ 4) 0
13. $\tanh^{-1}\left(\frac{1}{2}\right) =$
 1) $\frac{1}{2}\log_e 3$ 2) $\frac{1}{2}\log_e 2$
 3) $\log_e 3$ 4) $\log_e 5$
14. $\tanh^{-1}\left(\frac{1}{4}\right) + \coth^{-1}(4) =$
 1) $\frac{1}{2}\log_e\left(\frac{5}{3}\right)$ 2) $\log_e\left(\frac{5}{3}\right)$
 3) $\log_e(5)$ 4) $\log_e(3)$
15. $\operatorname{sech}^{-1}\left(\frac{1}{\sqrt{2}}\right) =$
 1) $\log_e(\sqrt{2} - 1)$ 2) $\log_e(\sqrt{2} + 1)$
 3) $\log_e(\sqrt{3} + 1)$ 4) $\log_e(\sqrt{3} - 1)$
16. If $\cosh^{-1}(x) = \log_e(2 + \sqrt{3})$ then $x =$
 1) 2 2) 3 3) 1 4) 5
17. If $|x| < 1$ then $\frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right) =$ (EAM-1999)
 1) $\tanh x$ 2) $\sinh^{-1}(x)$
 3) $\coth^{-1}(x)$ 4) $\tanh^{-1}(x)$
18. If $x = \log_e\left(y + \sqrt{y^2 - 1}\right)$ then $y =$
 1) $\tanh x$ 2) $\coth x$ 3) $\sinh x$ 4) $\cosh x$
19. $e^{\log_e(\cosh^{-1} 2)} =$
 1) $\log_e(2 - \sqrt{3})$ 2) $\log_e(3 - \sqrt{2})$
 3) $\log_e(\sqrt{3} + 2)$ 4) $\log_e(\sqrt{5} + 2)$
20. If $\log_e(\sqrt{5} + 2) = \sinh^{-1}(k)$ then $k =$
 1) 1 2) 5 3) 3 4) 2
21. $e^{\sinh^{-1}(\tan \theta)} =$
 1) $\sec \theta$ 2) $\tan \theta$
 3) $\sec \theta + \tan \theta$ 4) $\sec \theta - \tan \theta$
22. $\operatorname{sech}^{-1}\left(\frac{1}{3}\right) =$
 1) $\log_e(3 + \sqrt{8})$ 2) $\log_e(3 - \sqrt{8})$
 3) $\log_e(\sqrt{3} - \sqrt{8})$ 4) $\log_e(\sqrt{3} + \sqrt{8})$
23. If $u = \log_e\left[\tan\left(\frac{3\pi}{4} + \frac{\theta}{2}\right)\right]$ then $\cosh u =$
 1) $\sec \theta$ 2) $\tan \theta$ 3) $-\sec \theta$ 4) $-\tan \theta$
24. If $x > 0$ then $\tanh^{-1}\left(\frac{x^2 - 1}{x^2 + 1}\right) =$
 1) $\log_e(2x)$ 2) $\log_e x$
 3) $\log_e(3x)$ 4) $\log_e(5x)$
25. If $\cosh(3\theta) = 99$ then $\cosh \theta =$
 1) 3 2) 9 3) 11 4) 27
26. $\cosh(ix) =$
 1) $i \cosh x$ 2) $\cos x$
 3) $\cosh x$ 4) $-i \cosh x$

LEVEL-I-(H.W)-KEY

- 01) 3 02) 4 03) 1 04) 3 05) 1
 06) 2 07) 3 08) 1 09) 3 10) 2
 11) 1 12) 4 13) 1 14) 2 15) 2
 16) 1 17) 4 18) 4 19) 3 20) 4
 21) 3 22) 1 23) 3 24) 2 25) 1
 26) 2

LEVEL-I-(H.W)-HINTS

- The domain of $\cot hx$ is $R - \{0\}$
- The range of $\tanh x$ is $(-1, 1)$
- $\sinh(0) = \frac{e^0 - e^{-0}}{2}$
- $\cosh(1) - \sinh(1) = \frac{1}{e}$
- $(\cosh x - \sinh x)^n = e^{-nx}$
- $\cosh x = \frac{5}{3} \Rightarrow \sinh(x) = \pm \frac{4}{3}$
- $\cosh x = \frac{5}{2} \Rightarrow \cos 2x = 2 \cosh^2 x - 1$
- $\operatorname{sech} x = \frac{3}{5} \Rightarrow \tanh x = \pm \frac{4}{5}$
- $\cosh^4 2 - \sinh^4 2 = \cosh(4)$
- $\sinh 3x = 3 \sinh x + 4 \sinh^3 x$
- $\sinh^{-1}(2) = \log(2 + \sqrt{5})$
- $\cosh^{-1}(1) = \log(1) = 0$
- $\tanh^{-1}(x) = \frac{1}{2} \log\left(\frac{1+x}{1-x}\right)$
- $\tanh^{-1}(x) + \coth^{-1}\left(\frac{1}{x}\right) = 2 \tanh^{-1} x$
- $\operatorname{sech}^{-1} x = \log\left(\frac{1 + \sqrt{1-x^2}}{x}\right)$
- $\cosh^{-1} x = \log(x + \sqrt{x^2 - 1})$
- $\frac{1}{2} \log\left(\frac{1+x}{1-x}\right) = \tanh^{-1} x$

- $x = \log(y + \sqrt{y^2 - 1}) \Rightarrow y = \cosh x$
- $\cosh^{-1} 2 = \log(2 + \sqrt{3})$
- $\log(2 + \sqrt{2^2 + 1}) = \sinh^{-1}(2)$
- $e^{\log(\tan \theta + \sec \theta)} = \sec \theta + \tan \theta$
- $\operatorname{sech}^{-1} x = \log\left(\frac{1 + \sqrt{1-x^2}}{x}\right)$
- $e^u = \frac{-1 + \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}$
- $\tanh^{-1}\left(\frac{x^2 - 1}{x^2 + 1}\right) = \log x$
- $4 \cosh^3 x - 3 \cosh x = 99$
- $\cosh(ix) = \cos x$



LEVEL - II (C.W)

- The domain of $\cosh^{-1}(2x)$ is
 1) $(-\infty, \infty)$ 2) $[1, \infty)$
 3) $[2, \infty)$ 4) $\left[\frac{1}{2}, \infty\right)$
- The range of $\cosh\left(\frac{x}{2}\right)$ is
 1) R 2) $(0, \infty)$ 3) $(0, 1)$ 4) $[2, \infty)$
- The value of $\frac{\cosh 2\theta - 1}{\sinh 2\theta}$ is equal to
 1) $\cosh \theta$ 2) $\tanh \theta$ 3) $\operatorname{cosech} \theta$ 4) $\operatorname{sech} \theta$
- If $\tanh^2 x = \cos \theta$ then $\cosh 2x =$
 1) $\tan^2\left(\frac{\theta}{2}\right)$ 2) $\cot^2\left(\frac{\theta}{2}\right)$
 3) $\sec^2\left(\frac{\theta}{2}\right)$ 4) $\sin^2\left(\frac{\theta}{2}\right)$

5. If $\cosh x = \sec \theta$ then $\tanh^2 \left(\frac{x}{2} \right) =$

1) $\tan^2 \frac{\theta}{2}$ 2) $\cot^2 \frac{\theta}{2}$

3) $-\tan^2 \frac{\theta}{2}$ 4) $-\cot^2 \frac{\theta}{2}$

6. $\frac{\tanh x}{\operatorname{sech} x - 1} + \frac{\tanh x}{\operatorname{sech} x + 1} =$

1) $\operatorname{cosech} x$ 2) $2\operatorname{cosech} x$
3) $-\operatorname{cosech} x$ 4) $-2\operatorname{cosech} x$

7. If $\sin x \cosh y = \cos \theta, \cos x \sinh y = \sin \theta$
then $\sinh^2 y =$

1) $\cosh^2 x$ 2) $\cos^2 x$ 3) $\cosh^3 x$ 4) $\sinh^2 x$

8. If x is an acute angle and

$y = \log_e \left[\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right]$ then $\cos x \cdot \cosh y =$

1) 0 2) -1 3) 1 4) 2

9. $\sinh(\cosh^{-1} x) =$

1) $\sqrt{x^2 + 1}$ 2) $\frac{1}{\sqrt{x^2 + 1}}$

3) $\sqrt{x^2 - 1}$ 4) $\frac{1}{\sqrt{x^2 - 1}}$

10. $\log_e \left[(x-1) + \sqrt{x^2 - 2x} \right], x \geq 2$ is equal to

1) $\sinh^{-1}(x-1)$ 2) $\cosh^{-1}(x-1)$

3) $\sinh^{-1}(x+1)$ 4) $\cosh^{-1}(x+1)$

LEVEL - II (C.W)-KEY

01) 4 02) 4 03) 2 04) 2 05) 1
06) 4 07) 2 08) 3 09) 3 10) 2

LEVEL - II (C.W)-HINTS

1. The domain of $\cosh^{-1} 2x$ is $\left[\frac{1}{2}, \infty \right)$

2. The range of $\cosh x$ is $[1, \infty)$

3. $\frac{\cosh 2\theta - 1}{\sinh 2\theta} = \tanh \theta$

4. $\cosh 2x = \frac{1 + \tanh^2 x}{1 - \tanh^2 x}$

5. $\tanh^2 \left(\frac{x}{2} \right) = \frac{\cosh x - 1}{\cosh x + 1}$

6. $\frac{\tanh x}{\operatorname{sech} x - 1} + \frac{\tanh x}{\operatorname{sech} x + 1}$
 $= -2 \operatorname{cosech} x$

7. $\sin^2 x \cdot \cosh^2 y + \cos^2 x \cdot \sinh^2 y = 1$

$\Rightarrow (1 - \cos^2 x) \cosh^2 y + \cos^2 x (\cosh^2 y - 1) = 1$

8. $e^y = \frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}}$

9. $\sinh(\cosh^{-1} x) = \sqrt{x^2 - 1}$

10. $\log \left[(x-1) + \sqrt{(x-1)^2 - 1} \right] = \cosh^{-1}(x-1)$

LEVEL - II (H.W)

1. The domain of $\sec h^{-1} \left(\frac{x}{2} \right)$ is

1) $\left(0, \frac{1}{2} \right)$ 2) $\left[0, \frac{1}{2} \right]$ 3) $\left(0, \frac{1}{2} \right]$ 4) $(0, 2]$

2. $\frac{\sinh(2x)}{1 + \cosh(2x)} =$

1) $\tanh(x)$ 2) $\tanh^2(x)$

3) $\coth(x)$ 4) $\coth^2(x)$

3. $\cosh 2x + \sinh 2x =$

1) $\frac{1 + \tanh(x)}{1 - \tanh(x)}$ 2) $\frac{1 - \tanh x}{1 + \tanh x}$

3) $\frac{\tanh x - 1}{\tanh x + 1}$ 4) $\frac{1 - \tanh x}{\tanh x - 1}$

4. $\tanh^2(x) = \tan^2 \theta$ then $\cosh 2x =$

1) $\cos 2\theta$ 2) $\sec 2\theta$ 3) $\sin 2\theta$ 4) $\cos \theta$

5. $\frac{\cosh(x)}{1 - \tanh x} + \frac{\sinh(x)}{1 - \coth(x)} =$
 1) $\sinh(x) + \coth(x)$
 2) $\tanh(x) + \coth(x)$
 3) $\tanh(x) - \coth x$
 4) $\sinh(x) + \cosh(x)$
6. $\tan A = \tan \alpha \tanh \beta$ and
 $\tan B = \cot \alpha \tanh \beta$ then $\tan(A+B) =$
 1) $\sinh(2\beta) \cos 2\alpha$ 2) $\sinh(2\beta) \operatorname{cosec}(2\alpha)$
 3) $\cosh(2\beta) \sec 2\alpha$ 4) $\cosh(2\beta) \sin 2\alpha$
7. If $\sin x \sinh y = \cos \theta$ and
 $\cos x \cosh y = \sin \theta$ then $\cosh^2 y + \cos^2 x =$
 1) -1 2) 0 3) 1 4) 2
8. $\sinh^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) =$
 1) $\cosh^{-1} x$ 2) $\cosh^{-1}(2x)$
 3) $\tanh^{-1} x$ 4) $\tanh^{-1}(2x)$
9. $\operatorname{sech}^{-1}\left(\frac{2}{3}\right) = \log_e \left(\frac{(k+1) + \sqrt{k^2+1}}{k} \right)$ then
 $k =$
 1) 2 2) 3 3) 5 4) 6
10. For $0 < x \leq \pi$, $\sinh^{-1}(\cot x) =$ (Eam-2011)
 1) $\log_e \left(\cot \frac{x}{2} \right)$ 2) $\log_e \left(\tan \frac{x}{2} \right)$
 3) $\log_e (1 + \cot x)$ 4) $\log_e (1 + \tan x)$

LEVEL - II (H.W)-KEY

- 01) 4 02) 1 03) 1 04) 2 05) 4
 06) 2 07) 4 08) 3 09) 1 10) 1

LEVEL - II (H.W)-HINTS

1. The domain of $\operatorname{sech}^{-1} x$ is $(0, 1]$

2. $\frac{\sinh 2x}{1 + \cosh 2x} = \tanh x$
3. $\cosh 2x + \sinh 2x =$
 $= \frac{1 + \tanh^2 x}{1 - \tanh^2 x} + \frac{2 \tanh x}{1 - \tanh^2 x}$
4. $\cosh 2x = \frac{1 + \tanh^2 x}{1 - \tanh^2 x}$
5. $\frac{\cosh^2 x}{\cosh x - \sinh x} + \frac{\sinh^2 x}{\sinh x - \cosh x} =$
6. $\tan(A+B) = \frac{\tan \alpha \tanh \beta + \cot \alpha \tanh \beta}{1 - \tan \alpha \tanh \beta \cot \alpha \tanh \beta}$
7. $\sin^2 x \sinh^2 y + \cos^2 x \cosh^2 y = 1$
 $\Rightarrow (1 - \cos^2 x)(-1 + \cosh^2 y) + \cos^2 x \cosh^2 y = 1$
8. $\sinh^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) = \log\left(\frac{x+1}{\sqrt{1-x^2}}\right)$
 $= \frac{1}{2} \log\left(\frac{1+x}{1-x}\right)$
9. $\operatorname{sech}^{-1} x = \log\left(\frac{1 + \sqrt{1-x^2}}{x}\right)$
10. $\sinh^{-1}(\cot x) = \log(\cot x + \sqrt{\cot^2 x + 1})$



LEVEL - III

1. If $\sinh^3 x - \cosh^3 x = \frac{ke^x - e^{kx}}{1-k}$ then $k =$
 1) -1 2) 0 3) -3 4) 2
2. If $\tan \frac{x}{2} \coth\left(\frac{x}{2}\right) = 1$ then $\cos x \cosh x =$
 1) 1 2) 2 3) 0 4) -1

3. $2(\sinh 2 + \sinh 3 + \sinh 5) \frac{e^5}{e^5 - 1}$
 $= a + be^2 + ce^3 + de^5$ then $a+c+b+d=$
 1) 1 2) 2 3) 3 4) 4
4. $\sinh^{-1}(x) + \cosh^{-1}(x) = y$ then $\sinh(y) =$
 1) $x^2 + \sqrt{x^4 - 1}$ 2) $x^2 - \sqrt{x^4 - 1}$
 3) $x^2 + \sqrt{x^4 + 1}$ 4) $x^2 - \sqrt{x^4 + 1}$
5. $\sinh^{-1}(2\alpha) = 2 \cosh^{-1}(\beta)$ then
 1) $\alpha^2 + \beta^2 = \alpha^4$ 2) $\alpha^2 + \beta^2 = 4$
 3) $\alpha^2 + \beta^2 = \beta^4$ 4) $\alpha^2 = \beta^2$
6. The value of $\tan \left[i \log \left(\frac{a-ib}{a+ib} \right) \right]$ is
 1) $\frac{ab}{a^2 + b^2}$ 2) $\frac{ab}{a^2 - b^2}$ 3) $\frac{2ab}{a^2 - b^2}$ 4) $\frac{2ab}{a^2 + b^2}$
7. $\sec h^2 \left[\tanh^{-1} \left(\frac{1}{2} \right) \right] + \operatorname{cosec} h^2 (\coth^{-1}(3)) =$
 1) $\frac{35}{9}$ 2) $\frac{43}{4}$ 3) $\frac{35}{4}$ 4) $\frac{43}{9}$

LEVEL - III-KEY

- 1) 3 2) 1 3) 4 4) 1 5) 3
 6) 3 7) 3

LEVEL - III-HINTS

1. $\sinh^3 x - \cosh^3 x =$
 $(\sinh x - \cosh x)(\cosh^2 x + \sinh^2 x + \cosh x \sinh x)$
2. $\tan \frac{x}{2} = \tanh \frac{x}{2}$

$$\Rightarrow \cos x \cosh x = \left(\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right) (\cosh x)$$

$$= \operatorname{sech} x \cdot \cosh x$$

$$3. \quad 2 \left[\frac{e^2 - e^{-2} + e^3 - e^{-3} + e^5 - e^{-5}}{2} \right] \frac{e^5}{e^5 - 1}$$

$$= e^2 + e^3 + e^5 + 1$$

$$\therefore a=1, b=1, c=1, d=1$$

$$4. \quad \sinh^{-1} x + \cosh^{-1} x = y$$

$$y = \log(x + \sqrt{x^2 + 1})(x + \sqrt{x^2 - 1})$$

$$\Rightarrow e^y = (x + \sqrt{x^2 + 1})(x + \sqrt{x^2 - 1})$$

$$5. \quad \log(2\alpha + \sqrt{4\alpha^2 + 1}) = 2 \log(\beta + \sqrt{\beta^2 - 1})$$

$$\Rightarrow 2\alpha + \sqrt{4\alpha^2 + 1} = (\beta + \sqrt{\beta^2 - 1})^2$$

$$\text{after simplification } \alpha^2 + \beta^2 = \beta^4$$

$$6. \quad \text{By using the formula } \tan ix = i \tanh x$$

$$7. \quad \operatorname{sech}^2 \left(\tanh^{-1} \frac{1}{2} \right) + \operatorname{cosec} h^2 (\coth^{-1} 3)$$

$$= 1 - \tanh^2 \left(\tanh^{-1} \frac{1}{2} \right) + \coth^2 (\coth^{-1} 3) - 1$$